

Unified Learning Portal using AngularJS

1. Introduction

The Unified Learning Portal is a web-based application developed using AngularJS. It demonstrates the concept of a Single Page Application (SPA) where different sections of the application are dynamically loaded without refreshing the entire webpage. The system is designed for BTech Computer Engineering students to support their study of AngularJS across five units. It allows users to explore the course syllabus unit-wise, track lab practicals with completion status and personal notes, and browse reference textbooks. AngularJS routing is used to navigate between pages such as Overview, Practical, and References, while features like two-way data binding, custom directives, filters, form validation, and HTTP services are actively used throughout the application itself — making it both a learning tool and a live demonstration of the concepts it teaches.

2. Objectives

- Use ngRoute for client-side navigation between views.
- Display course syllabus unit-wise for BTech Computer Engineering students.
- Track lab practical completion with notes and progress indicators.
- Fetch and display reference textbooks using \$http and \$q services.
- Build a clean, user-friendly academic web interface.

3. Technologies Used

Technology	Description
HTML	Used to structure the web pages
CSS	Used to design and style the interface
JavaScript	Used to implement application logic
AngularJS	Used to build the Single Page Application
ngRoute	Used to handle routing between pages
\$http / \$q	Used to fetch data from JSON files asynchronously
Custom Directives	Used to create reusable UI components (status Card)
JSON	Used to store and serve textbook reference data

4. System Features

- Overview Page – Shows course syllabus unit-wise with topics and study material.
- Lab Tracker – Track completion of 30 practicals with notes and a progress bar.
- References Section – Fetches and displays textbook data from a JSON file.
- Submission Form – Submit assignments with AngularJS form validation.
- SPA Navigation – Seamless routing between pages using ngRoute.
- Dynamic Data Binding – View updates automatically with scope changes.

5. System Architecture

- index.html – Main file that loads AngularJS and hosts the ng-view outlet.
- app.js – Configures routing using \$routeProvider for all three views.
- controllers/ – Four controllers handling Overview, Lab, References, and Submission logic.
- services/dataService.js – Fetches JSON data using \$http and \$q.
- directives/statusCard.js – Custom reusable directive for practical status cards.
- views/ – Three HTML templates for Overview, Practicals, and References pages.
- css/style.css – Styles the entire application.
- data/textbooks.json – Stores reference textbook data.

6. Working of the System

- When the application loads, the Overview page is displayed with course syllabus and student info.
- Users can navigate between Overview, Practicals, and References using the navigation menu.
- The syllabus section displays all five AngularJS units with topics and study material.
- Users can search and sort units dynamically using AngularJS filters.
- On the Practicals page, users can mark practicals as complete and add personal notes.
- A progress bar updates automatically as practicals are completed.
- The References page fetches and displays textbook data from a JSON file.
- Users can submit assignments via the form with real-time validation feedback.
- AngularJS updates the view dynamically without reloading the page.

7. Advantages

- Simple and clean interface suitable for academic use.
- No page refresh required due to SPA architecture.
- Covers all core AngularJS concepts in a single project.
- Helps track lab practical completion with progress indicator.
- Demonstrates real-world async data fetching using \$http and \$q.
- Useful for learning and revising the BTech AngularJS syllabus.

8. Limitations

- Not optimized for mobile devices.
- Textbook data is hardcoded in a static JSON file — cannot be updated dynamically.

9. Future Enhancements

- Add backend (Node.js + MongoDB) for permanent data storage.
- Implement user login and authentication.
- Migrate to Angular (v17+) or react for better performance.
- Add a quiz module for unit-wise self-assessment.
- Enable dark mode and mobile responsiveness.
- Add code syntax highlighting using Highlight.js.
- Allow PDF export of notes and progress report.

10. Conclusion

The Unified Learning Portal using AngularJS successfully demonstrates the implementation of a Single Page Application with client-side routing, two-way data binding, custom directives, form validation, and asynchronous data fetching. The project covers all five units of the BTech Computer Engineering AngularJS syllabus and provides students with an interactive platform to explore course content, track lab practical completion, and submit assignments. It serves as both a functional academic tool and a live demonstration of core AngularJS concepts, making it a valuable learning project for frontend web development.